

COMPACT MULTI-PURPOSE IMAGER FOR THE MATTER IN EXTREME CONDITION END-STATION AT LCLS

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Abstract

We present the mechanical design of a new imaging system developed at the Matter in Extreme Conditions (MEC) instrument at the Linac Coherent Light Source (LCLS) to improve setup efficiency while maintaining high-quality imaging performance. We designed an in-vacuum setup for imaging the focal spot of the different laser systems we have at MEC as well as the experimental targets themselves. The system integrates high-resolution optics, remote positioning relative to the interaction point and control system. It supports imaging of laser spot sizes ranging from 2 to 600 μm and offers spatial resolutions down to 1 μm . Using kinematic mounting features, we ensure accuracy of the alignment of internal optics and allow for user-friendly modifications when needed.

Multiple magnification and resolution options are supported to cover a wide range of focal conditions. Additionally, the system includes vacuum-compatible, adjustable wavelength filtering and attenuation that maintain optical alignment. Finally, a shutter protects the high-resolution optics from target debris while the whole imager is fully retractable to further clear the target area.

INTRODUCTION

The Matter in Extreme Conditions (MEC) instrument at the Linac Coherent Light Source (LCLS) [1] is designed to investigate a wide range of phenomena in high energy density physics. The experimental requirements are diverse, necessitating imaging of both targets and laser beams across a broad range of spatial scales. For example, the high-energy Nd:glass laser system, typically employed for shock physics [2], produces large focal spots requiring characterization up to 600 μm in diameter. In contrast, the Ti:Sapphire laser, used for studies of warm dense matter [3] and relativistic plasmas [4], demands measurements of spot sizes as small as 2 μm . In addition, accurate imaging of experimental targets is required with varying resolution and field-of-view specifications, along with precise alignment of optical and x-ray beams [5]. These needs must be met within a 2 m-diameter vacuum vessel, where the experimental configuration changes monthly. To address these challenges, a versatile, user-friendly, and geometrically compatible imaging system is essential. The newly commissioned Multi-Purpose Imager (MPI) has been developed to fulfill this role.

The MEC experimental chamber is located about 460 m from the end of the LCLS undulator. The large volume of the chamber provides room for multiple vacuum-compatible spectrometers and scattering diagnostics instruments. The MEC laser beam is routed to the interaction region in

multiple configurations. As a result, the MEC end-station is versatile with configurations changed frequently for different experiments, requiring precise, repeatable positioning and alignment.

This paper presents the optomechanical design of a new in-vacuum imaging system aimed at improving experimental set up efficiency while maintaining high-quality / high resolution imaging of the laser spot as well as the targets themselves.

Key Requirements

- A compact retractable package compatible with 0.1 Pa (10^{-5} mbar) experimental target chamber vacuum levels.
- High resolution imaging of the laser focal spot and target for precise positioning.
- Rapid removal and reinstallation with minimum realignment required.
- Support of multiple magnification/resolution options.
- Provide multiple adjustable filters (wavelength and attenuation).

In addition, the instrument needs to include a shutter to protect optics and detectors from debris and accidental full power laser hits

DESIGN

Figure 1 shows the optical layout. The design allows imaging of targets (with the fiber light illuminator) as well as the focused pump laser (spot sizes from $\sim 3 \mu\text{m}$ to $\sim 500 \mu\text{m}$).

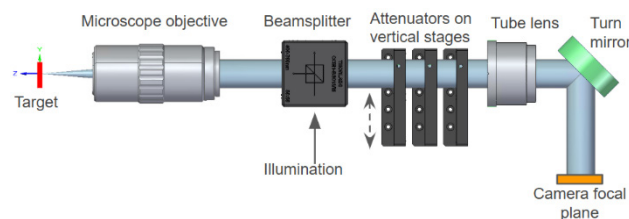


Figure 1: Optical layout.

Layout consists of an infinity-corrected long working distance microscope objective (e.g. Mitutoyo APO) plus a tube lens. The microscope is compatible with 95 mm parfocal length, infinity corrected, visible light microscope objectives with working distance $\geq 15 \text{ mm}$ (i.e. Mitutoyo 2x, 5x, 10x and 20x). The filters are placed into the collimated portion of the beam path so that the image location is not affected. Six attenuators are provided allowing attenuation of OD 0 to OD 10 in steps of 0.5.

Mechanical Design

To satisfy the requirements for motorized retractability, focus and, lateral alignment, the XYZ stage stack is placed at the bottom of the MPI body. The cantilever mounting allows the instrument to reach closely to the interaction point (IP) without interfering with other diagnostics instruments. Figure 2 shows a CAD model of the MPI. The center of gravity is over the base plate to avoid tipping over when the MPI is free standing. Vacuum compatible plastic, Polyoxymethylene, pads, mounted on the bottom of the base plate, protect the chamber optical table from scratching when the instrument is repositioned. The patch panel facilitates quick installation and removal. During installation the alignment plate is pressed against the base of the central hexapod, providing the appropriate initial spacing between the object/sample and objective. Angular positioning is achieved by rotating the MPI around the hexapod base.

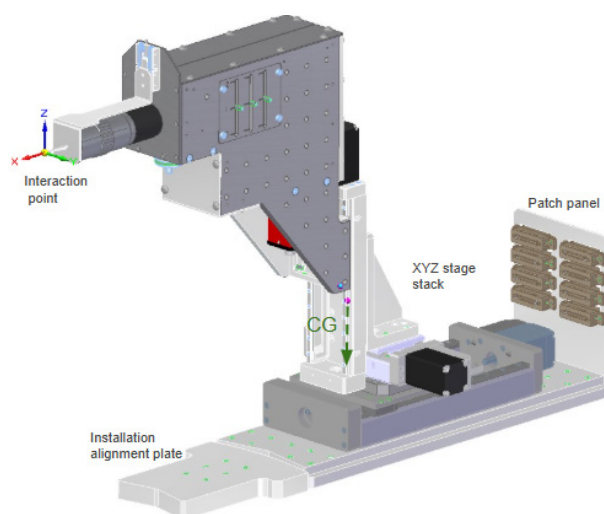


Figure 2: Multi-Purpose Imager CAD model.

The main body of the imager is constructed of two parallel aluminum plates that serve as a structural frame. As shown in Fig. 3, all optical elements are mounted onto one plate, allowing for precise positioning and ease of installation and servicing. For stability, the major optical components are not adjustable and aligned by mechanical design only.

Semi kinematic mounting features precisely position the holders of optical elements allowing for quick, easy, and repeatable positioning and servicing. The aluminum plate has machined recesses where the components' mounts are secured. The referencing features are designed to work with gravity, eliminating user error during assembly.

The internal surfaces are anodized to minimize image degradation due to internal reflections.

The 90° turn mirror improves the packaging of the MPI, avoiding interference with other instruments. The attenuators are placed on three vertical Piezo stages allowing for

motorized selection of different optical density filter combinations.

Figure 4 shows the assembled instrument in the MEC chamber. One of the side thick Aluminum plates is removed to show internal components.

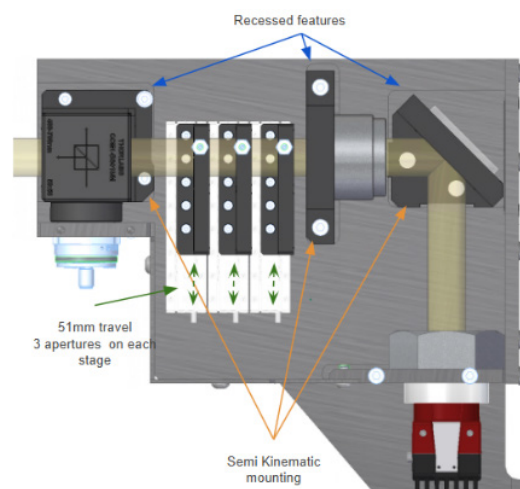


Figure 3: Mounting of optical components.



Figure 4: Multipurpose image in MEC chamber.

IMAGING

The spatial resolution of the imaging system was evaluated using a standard USAF 1951 resolution test target. For this characterization, we employed an Alvium 1800 U-1236 optical camera with a detector array of 4122×3008 pixels. The imaging optics consisted of a 20x Mitutoyo long-working-distance objective lens with a numerical aperture (NA) of 0.28. Under these conditions, the system was able to clearly resolve features corresponding to Group 9, Element 3 of the resolution target, which corresponds to a line width of $0.78 \mu\text{m}$. This level of performance, see Fig. 5, demonstrates that the

optical configuration provides sub-micrometer resolving capability, consistent with expectations based on the numerical aperture and magnification of the objective.

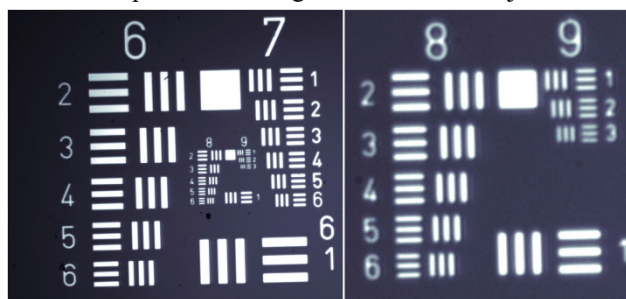


Figure 5: Image of USAF 1951 resolution Target. Left: Groups 6 to 9. Right: zoom of group 8 and group 9. Group 9 element 3 has a linewidth of $0.78\ \mu\text{m}$ and is still resolved.

CONCLUSION

The MPI is a compact, retractable, in-vacuum setup for imaging both the focal spot of a laser and the targets themselves. The optical and mechanical design emphasizes stability and simplicity during assembly and alignment. The new instrument has been installed in the chamber and will be used in the next experimental run in MEC at LCLS.

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